Palash Gharde

Game Developer / Technical Gameplay Designer

(602) 565 - 9146 | San Francisco, CA | palash.gharde047@gmail.com | linkedin.com/in/palashgharde/ | Portfolio | Playable Demos

SUMMARY

Game Developer / Technical Gameplay Designer with experience building modular gameplay systems, AI behavior systems, multiplayer mechanics, and rapid prototypes in **Unreal Engine 5** and **Unity**. Strong software engineering foundation with experience translating backend development and scalable system design into modular gameplay architecture and player-focused systems.

TECHNICAL SKILLS

Game Engines	:	Unreal Engine 5 (UE5), Unity
Programming	:	Blueprints, C#, Python, C++, JavaScript, Object-Oriented Programming (OOP), Event-driven architecture, Problem-solving, Debugging
Gameplay Systems	:	Interaction systems, Player controllers, Inventory systems, AI Behavior Trees, AI Perception, Economy systems, Procedural generation
Multiplayer	:	Unity Netcode for GameObjects, host-client synchronization
Design & Production	:	Game design documentation, Level Design, Gameplay Prototyping, Rapid Iteration, Feature Planning
Tools	:	Git, GitHub, Visual Studio, HacknPlan, Miro, Figma, Photoshop, Blender

PROJECTS

Kidnapped – *Unreal Engine 5* | *Portfolio* | *Itch.io*

First-person survival puzzle game focused on stealth, exploration, and escape-based gameplay

- Solo-developed a complete UE5 playable build over 5 months, owning gameplay mechanics, level design, sound effects, basic Blender assets, cutscenes, and final integration.
- Constructed modular interaction framework with **Blueprint Interfaces** and parent/child Blueprint hierarchy, supporting 20+ interactable object types without duplicated logic.
- Designed and integrated **survival** and **progression systems** including hunger, thirst, stamina, and inventory systems.
- Engineered **NPC AI** using **Behavior Trees** and **AI Perception** with patrol, investigate, chase, and reset behavior states.
- Integrated **Level Sequencer** driven narrative events into a structured 20-30 minute play session across 8 puzzle sequences.

The Elemental Ascent – *Unreal Engine 5* | *Portfolio*

Third-person puzzle-platformer centered around physics-based elemental interactions

- Independently **authored GDD** defining elemental interaction rules, core loop, puzzles, controls, and production scope.
- Prototyped physics-based elemental mechanics in Blueprints and iterated puzzle clarity, difficulty pacing, and spatial readability through playtesting.
- Configured **UE5 Enhanced Input** for keyboard/mouse and controller, improving consistency across movement, camera, and ability interactions.
- Converted 2D layouts into **3D blockouts** and refined encounter flow, puzzle readability, and traversal timing across multiple iteration cycles.

Burger Shop Simulator – *Unity C#* | *GitHub*

2.5D top-down multiplayer restaurant management and cooking simulation game in Unity

- Solo-developed player interaction, pickup/drop logic, plate assembly, and order delivery systems in Unity C#.
- Built an **inheritance-based** counter architecture with a shared BaseCounter class, overriding interaction logic for cutting, stove, ingredient container, delivery, and trash counters.
- Used **event-based communication** to decouple kitchen objects, recipe tracking, UI updates, and delivery flow, improving maintainability and expansion potential.
- Implemented **multiplayer** functionality using **Unity Netcode** for GameObjects, synchronizing player actions, object pickup/drop, counter interactions, recipe progress, and shared gameplay state across host/client sessions.

Jetpackman Adventures – *Unity C#* | *Portfolio* | *Itch.io*

2D flying game with procedural obstacle generation and minigames

- Programmed boost-based flying movement, collision handling, scoring, pause/resume flow, and game-over state management for PC and mobile input.
- Created dual-weapon system: impact bombs destroy wooden doors while hacking lasers trigger a lock-hacking minigame with randomized pin placement.
- Built **procedural obstacle spawning** with randomized height variance, dynamic spawn timing, and difficulty scaling based on player progression.

Additional Projects

- **Hunting Simulator** - UE5 first-person hunting loop with animal AI, weapon upgrades, economy, and save progression
- **Wrong Way** - UE5 mobile endless driving with procedural track generation, powerups, coins, and randomized traffic system.

EXPERIENCE

Associate Software Engineer | ***Veritas Technologies LLC, Pune, India*** Jan 2020 – April 2022

- **Engineered scalable backend systems** and **automation tools** using **Python, Django, and AWS**, reducing manual testing effort by 40% and issue resolution time by 30%.
- Owned end-to-end SpecSFS benchmark **automation pipeline** for environment setup, filesystem setup, test execution, output parsing, and automated performance reporting.
- **Collaborated** in a 6-member Agile team to debug production issues, improve system reliability, and deliver maintainable software in iterative release cycles.

EDUCATION & CERTIFICATIONS

Arizona State University Tempe, AZ
Master of Science in Computer Science | GPA: 3.9/4.0 *August 2023 – May 2025*

Vishwakarma Institute of Technology Pune, India
Bachelor of Technology in Computer Engineering | GPA: 8.62/10.0 *August 2016 – May 2020*

CERTIFICATIONS

- Game Design with Unreal Engine 5 - April 2026